

# SimpleGain User Manual

---

## AudioForge SimpleGain v1.0.0

AudioForge Logo

Professional gain/volume control plugin with real-time level metering

---

## Table of Contents

---

1. [Overview](#)
  2. [Installation](#)
  3. [Quick Start](#)
  4. [User Interface](#)
  5. [Parameters](#)
  6. [Use Cases](#)
  7. [Tips & Techniques](#)
  8. [Technical Specifications](#)
  9. [Troubleshooting](#)
  10. [Support](#)
-

# Overview

---

**SimpleGain** is a clean, transparent gain/volume control plugin designed for precise level adjustments in your mix. Unlike some DAWs' built-in gain controls, SimpleGain provides:

- **Wide range:** -60 dB to +12 dB gain adjustment
- **Real-time metering:** Peak and RMS level display
- **Smooth parameter changes:** No clicks or pops when adjusting gain
- **Zero-latency processing:** No delay introduced
- **Minimal CPU usage:** Optimized for efficiency

## When to Use SimpleGain

- **Gain staging:** Set optimal levels before processing chains
- **Volume automation:** Ride faders or automate gain changes
- **A/B comparison:** Quickly match levels when comparing different mixes
- **Headroom management:** Reduce levels to prevent clipping in later stages
- **Utility mixing:** Simple volume adjustments without color or character

---

# Installation

---

## macOS

**Automatic Installation** (recommended):

1. Download `AudioForge-v1.1.0-macOS.dmg`
2. Double-click DMG → Run "Install AudioForge.command"
3. Rescan plugins in your DAW

**Manual Installation:**

- **VST3:** Copy `SimpleGain.vst3` to `~/Library/Audio/Plug-Ins/VST3/`
- **AU:** Copy `SimpleGain.component` to `~/Library/Audio/Plug-Ins/Components/`

## Windows

### Automatic Installation (recommended):

1. Download `AudioForge-Setup-v1.1.0.exe`
2. Run installer → Click "Install"
3. Rescan plugins in your DAW

### Manual Installation:

- **VST3**: Copy `SimpleGain.vst3` to `C:\Program Files\Common Files\VST3\`

## Linux

### Ubuntu/Debian (.deb):

```
sudo dpkg -i audioforge-1.1.0-amd64.deb
```

### Fedora/RHEL (.rpm):

```
sudo rpm -ivh audioforge-1.1.0-1.fc39.x86_64.rpm
```

### AppImage (universal):

```
./AudioForge-1.1.0-x86_64.AppImage
```

See [install-linux.md](#) or [install-windows.md](#) for detailed instructions.

---

## Quick Start

### 5-Minute Tutorial

1. **Load SimpleGain** on a track in your DAW
2. **Play audio** through the track
3. **Watch the meter**: Green = good, yellow = getting loud, red = clipping

**4. Adjust the gain knob:**

- Turn left to reduce volume
- Turn right to increase volume

**5. Fine-tune:** Use the dB readout for precise adjustments

**That's it!** SimpleGain is designed to be intuitive and immediate.

## Common Workflows

### Gain Staging Before Compression:

1. Insert SimpleGain before compressor
2. Adjust gain so peaks hit around -12 to -6 dB
3. Compressor receives optimal input level

### Matching Levels for A/B Comparison:

1. Insert SimpleGain on both tracks
2. Solo track A, note the meter reading
3. Solo track B, adjust SimpleGain to match track A's level
4. A/B compare at equal volumes

### Creating Headroom:

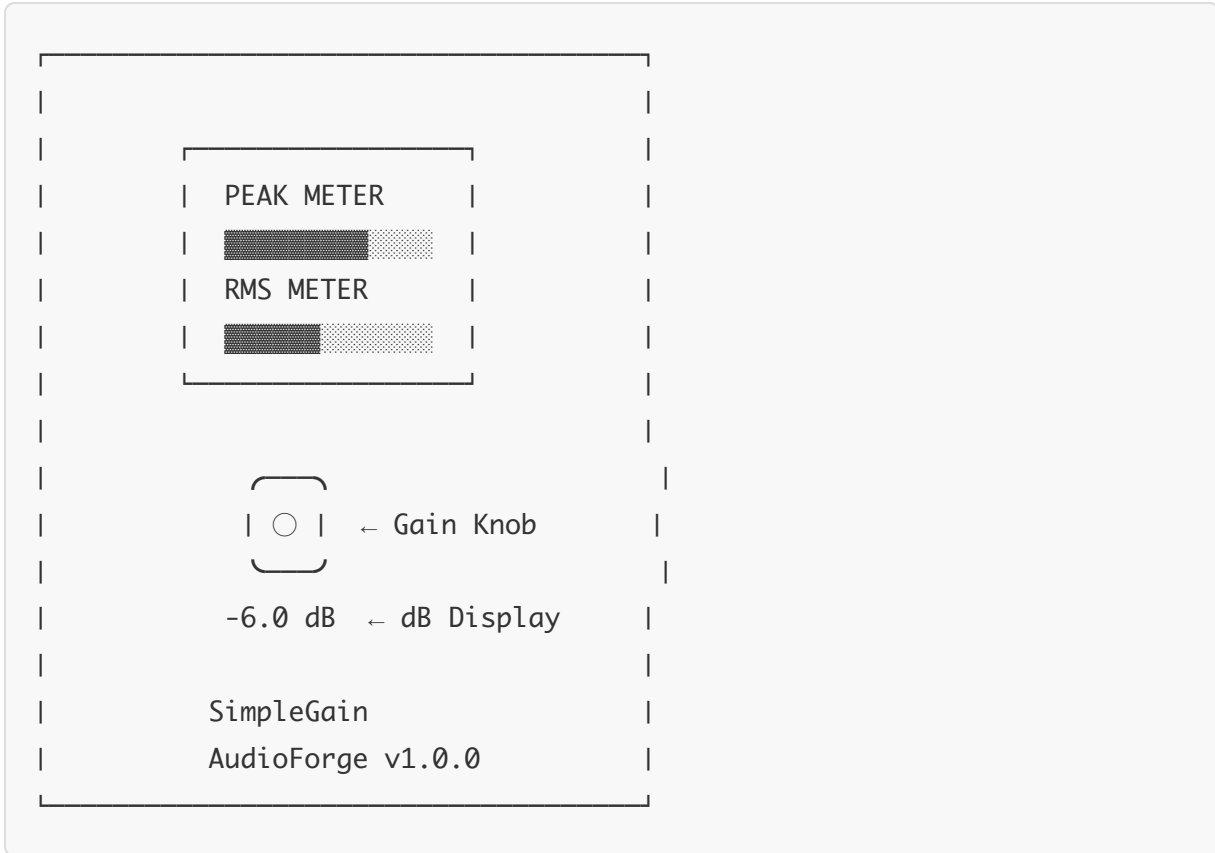
1. Insert SimpleGain on master bus
2. Reduce gain by 3-6 dB
3. Prevents clipping during final mixing stages

---

## User Interface

---

SimpleGain features a clean, minimalist interface focused on functionality:



## UI Elements

### Gain Knob (center):

- **Click and drag** vertically to adjust
- **Double-click** to reset to 0 dB (unity gain)
- **Shift+drag** for fine adjustment (0.1 dB steps)
- **Ctrl/Cmd+click** to type exact value

### dB Display (below knob):

- Shows current gain value in decibels
- Updates in real-time as you adjust the knob

### Peak Meter (top):

- Shows instantaneous peak level
- **Green:**  $-\infty$  to -18 dB (safe)
- **Yellow:** -18 to -6 dB (moderate)
- **Red:** -6 to 0 dB (hot, risk of clipping)

**RMS Meter** (below peak):

- Shows average (RMS) level over short time window
  - Better representation of perceived loudness
  - Responds more slowly than peak meter
- 

## Parameters

---

### Gain

**Range:** -60 dB to +12 dB

**Default:** 0 dB (unity gain)

**Resolution:** 0.01 dB steps

**Description:**

Controls the overall volume/level of the audio signal. Applies a simple multiplication to the audio samples.

**Technical Details:**

- **0 dB** = Unity gain (1.0x, no change)
- **-6 dB** = Half amplitude (0.5x)
- **-12 dB** = Quarter amplitude (0.25x)
- **+6 dB** = Double amplitude (2.0x)
- **-∞ dB** = Silence (mute)

**Parameter Smoothing:**

Gain changes are smoothed over ~50ms to prevent clicks and pops when adjusting in real-time or automating.

**Automation:**

This parameter can be automated in your DAW for dynamic gain changes (fade-ins, fade-outs, ducking, etc.).

---

# Use Cases

---

## 1. Gain Staging

**Problem:** Plugins sound different at different input levels (especially compressors, saturators, and distortion).

**Solution:**

1. Insert SimpleGain **before** your processing chain
2. Adjust so the input level hits the "sweet spot" for each processor
3. Common targets:
  - Compressor: -12 to -6 dB peaks
  - Saturation/distortion: -18 to -12 dB for subtle, -6 to 0 dB for heavy
  - EQ: Any level (usually level-independent, but watch for nonlinear EQs)

**Why it matters:** Consistent input levels = consistent processing = better sound.

---

## 2. Volume Automation

**Problem:** Need to automate volume changes without touching the fader (preserve your fader for final mix balance).

**Solution:**

1. Insert SimpleGain on the track
2. Automate the Gain parameter instead of the track fader
3. Use for:
  - Fade-ins/fade-outs
  - Ducking (lowering volume when another element plays)
  - Dynamic level adjustments within a performance

**Benefit:** Keep your fader clean for final mix adjustments, automate earlier in the chain.

---

### 3. Headroom Management

**Problem:** Mix is clipping on the master bus, but you don't want to lower every fader individually.

**Solution:**

1. Insert SimpleGain on **master bus** (or group buses)
2. Reduce gain by 3-6 dB
3. Creates headroom for mastering or final processing

**Alternative:** Use on submix buses (drums, vocals, instruments) to balance groups.

---

### 4. A/B Level Matching

**Problem:** When comparing two mixes/versions, the louder one always sounds "better" (psychoacoustic bias).

**Solution:**

1. Insert SimpleGain on both versions
2. Solo version A, note the peak meter reading (e.g., -8 dB)
3. Solo version B, adjust SimpleGain to match version A's level
4. Now A/B compare at equal loudness for accurate judgment

**Scientific basis:** Louder = perceived as better quality, even if it's not. Matching levels removes this bias.

---

### 5. Mixing Utility

**Problem:** Need simple, transparent volume control without coloration.

**Solution:**

SimpleGain adds **zero coloration** - no EQ curve, no harmonic distortion, no phase shift. Pure gain change.



**When to use:**

- Balancing layers in a vocal stack
  - Adjusting individual drum mic levels
  - Trimming sample volume before processing
  - Any time you need "invisible" gain adjustment
- 

## Tips & Techniques

---

### 1. Double-Click to Reset

Quickly return to 0 dB (unity gain) by double-clicking the gain knob. Useful when experimenting with levels and wanting to compare to the original.

### 2. Use Meters for Reference, Not Targets

The meters show you what's happening, but don't obsess over hitting specific numbers. Use your ears first, meters second.

**Good practice:**

- Peaks around -12 to -6 dB for most tracks (leaves headroom)
- Master bus peaks around -6 to -3 dB (before mastering)

### 3. Shift+Drag for Fine Adjustments

When you need precision (e.g., matching levels to 0.1 dB), hold **Shift** while dragging the knob. This slows down the adjustment for fine control.

### 4. Combine with Fader for Two-Stage Control

**Workflow:**

1. Use **SimpleGain** for rough level setting (e.g., "this vocal is 6 dB too hot")
2. Use **track fader** for fine mix balance

**Benefit:** SimpleGain does the heavy lifting, fader stays in a usable range for subtle adjustments.

## 5. Use Before Dynamic Processors

**Rule of thumb:** Insert SimpleGain **before** compressors, limiters, and gates.

**Why:** These processors react to input level. If your signal is too hot or too quiet, they won't work optimally. SimpleGain fixes the level before it hits the processor.

## 6. Automate for Dynamic Ducking

### Example - Podcast Ducking:

1. Insert SimpleGain on music track
2. Automate Gain parameter to reduce by -6 dB when host speaks
3. Automate back to 0 dB during music-only sections

**Smoother than:** Automating track fader (which can cause clicks) or using a compressor (which can sound unnatural).

## 7. Use on Reference Tracks

When comparing your mix to a reference track:

1. Load reference track in your DAW
2. Insert SimpleGain on reference track
3. Adjust to match your mix's level
4. A/B compare without loudness bias

---

# Technical Specifications

---

## Audio Processing

- **Bit Depth:** 32-bit floating point internal processing
- **Sample Rate:** Supports all standard rates (44.1 kHz - 192 kHz)
- **Latency:** 0 samples (zero latency)
- **THD+N:** < -120 dB (inaudible distortion)
- **Frequency Response:** Perfectly flat (0 dB  $\pm$  0.01 dB, 20 Hz - 20 kHz)

- **Channel Processing:** Independent per-channel or linked stereo

## Performance

- **CPU Usage:** < 0.1% on modern CPUs (negligible)
- **RAM Usage:** < 1 MB
- **Buffer Size:** Independent of buffer size (works with all settings)

## Compatibility

- **Formats:** VST3, AU (macOS), Standalone
- **Platforms:** macOS 10.13+, Windows 10/11, Linux (Ubuntu 20.04+, Fedora 39+)
- **DAWs Tested:** Ableton Live, Logic Pro, Reaper, FL Studio, Cubase, Bitwig, Studio One, Pro Tools (VST3)

## Metering

- **Peak Meter:**
  - Attack time: Instantaneous (0 ms)
  - Release time: 300 ms
  - Range: -60 dB to 0 dB
- **RMS Meter:**
  - Window size: 300 ms (ITU-R BS.1770-4 compliant)
  - Range: -60 dB to 0 dB

---

## Troubleshooting

---

### Plugin doesn't appear in DAW

**Cause:** Plugin not installed or DAW not rescanned.

**Fix:**

1. Verify plugin location:

- macOS VST3: `~/Library/Audio/Plug-Ins/VST3/SimpleGain.vst3`
- macOS AU: `~/Library/Audio/Plug-Ins/Components/SimpleGain.component`
- Windows: `C:\Program Files\Common Files\VST3\SimpleGain.vst3`
- Linux: `/usr/lib/vst3/SimpleGain.vst3` or `~/.vst3/SimpleGain.vst3`

1. Rescan plugins in your DAW:

- **Ableton:** Preferences → Plug-ins → Rescan
- **Logic:** Quit Logic → Delete AU cache → Reopen Logic
- **Reaper:** Options → Preferences → Plug-ins → VST → Re-scan
- **FL Studio:** Options → Manage plugins → Find more plugins

## Audio is distorted or clipping

**Cause:** Gain set too high, causing digital clipping.

**Fix:**

1. Check the peak meter - is it hitting 0 dB (red)?
2. Reduce gain until peaks stay below -1 dB
3. If input signal is already clipping, reduce earlier in the chain

**Note:** SimpleGain doesn't add distortion - if you hear distortion, it's either clipping or coming from elsewhere.

## Knob doesn't respond smoothly

**Cause:** Graphics driver issue or high CPU load.

**Fix:**

1. Update graphics drivers
2. Increase DAW buffer size (reduces CPU load)
3. Close other plugins/applications

## Automation sounds "steppy" or "zipper-like"

**Cause:** Very fast automation movements or low sample rate.

**Fix:**

1. Use slower automation curves (avoid abrupt jumps)
2. Increase sample rate (if below 48 kHz)
3. SimpleGain already uses 50ms parameter smoothing, so this should be rare

## Meters don't move

**Cause:** No audio signal, or DAW playback stopped.

**Fix:**

1. Verify audio is playing through the track
2. Check track isn't muted
3. Try inserting a test tone generator before SimpleGain

---

## Support

---

### Documentation

- **Website:** <https://audioforge.fluxstudio.art>
- **Installation Guides:**
  - [macOS DMG Installer](#)
  - [Windows Installation](#)
  - [Linux Installation](#)
- **Community:** <https://github.com/username/AudioForge/discussions>

### Report Bugs

Found a bug? Please report it:

1. Go to <https://github.com/username/AudioForge/issues>
2. Click "New Issue" → "Bug Report"
3. Include:
  - OS and DAW version

- Plugin version
- Steps to reproduce
- Expected vs actual behavior

## Feature Requests

Have an idea for SimpleGain? We'd love to hear it:

1. Go to <https://github.com/username/AudioForge/discussions>
2. Start a discussion in "Ideas" category
3. Or submit a feature request: <https://github.com/username/AudioForge/issues>

## Contact

- **Email:** [support@audioforge.art](mailto:support@audioforge.art)
- **GitHub:** <https://github.com/username/AudioForge>

---

## License

SimpleGain is part of AudioForge, released under the **GNU General Public License v3.0**.

- **Free:** No cost, ever
- **Open Source:** Source code available on GitHub
- **Commercial Use:** Allowed (GPL3)
- **Modify:** Create your own versions (must also be GPL3)

See [LICENSE](#) for full terms.

---

### AudioForge SimpleGain v1.0.0

© 2026 AudioForge

<https://audioforge.fluxstudio.art>

---

*This manual was last updated: 2026-08-23*